

Predicting Fatal Crash Outcomes

A Machine Learning Analysis of 2023 U.S. Traffic Fatalities

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Table of Contents

1. Executive Summary	3
2. Problem Statement	4
3. Data Source	4
4. Data Investigation	5
5. Data Cleansing	6
6. Feature Selection	7
7. Model Development and Evaluation	8
8. Practical Implications	10-11
9. Conclusion and Recommendations	12
References	13
Appendix	14-21

1. Executive Summary

Every year, thousands of families in the United States lose someone in a traffic crash. In 2023 alone, NHTSA recorded 37,654 fatal crashes across the country, and a large portion of those involved conditions that could have been anticipated. We wanted to understand what actually makes a crash more deadly, specifically whether we could predict the number of lives lost per incident using information available at the time of the crash. If we can, then transportation agencies, public health departments, and rideshare companies have a real data-backed starting point for where to focus resources.

Using the 2023 NHTSA Fatality Analysis Reporting System dataset, we merged two data files to build a dataset of 37,654 crashes with 14 predictors. Our target variable was FATALS, the number of fatalities per crash, which ranged from 1 to 8. About 93% of crashes resulted in exactly one death, which means the distribution is heavily skewed and predicting the rare multi-fatality crashes is the real challenge. We treated this as a regression problem and built two models: a Multiple Linear Regression model with stepwise feature selection, and a pruned Decision Tree.

Both models were trained on 70% of the data and validated on the remaining 30%. Before modeling, we handled NHTSA's coded unknown values, imputed missing data using the median and mode, capped outliers using the IQR method, and converted categorical variables to factors in R. We used stepwise AIC selection to narrow 80 original columns down to 10 final predictors, and we checked for multicollinearity using Variance Inflation Factors. The Linear Regression achieved a validation RMSE of 0.3467 and an MAE of 0.1505. The Decision Tree came in at 0.3457 and 0.1416. Five-fold cross-validation confirmed that neither model was overfitting.

The most influential predictors across both models were the number of persons involved, the number of vehicles, whether an impaired driver was present, lighting conditions, and road functional classification. These findings connect directly to the recommendations: the methods we used are not just academic exercises. They point to real patterns in when and where fatal crashes are most severe, and those patterns can be acted on.

What the models show, when you put it all together, is that the deadliest crashes are not random. They tend to happen at night on rural roads, involve impaired drivers, and include more vehicles and people. Linear regression lets us quantify exactly how much each of those factors contributes, and the decision tree confirms the same structure through a completely different approach. The fact that both models agree gives us more confidence that these findings are real and not just an artifact of one particular method. Based on that, we recommend that state agencies prioritize rural road lighting, that rideshare companies increase late-night driver availability on weekends, and that breathalyzer kiosk programs target high-volume bar areas in counties with limited transportation alternatives.

2. Problem Statement

Traffic fatalities in the United States have stayed high despite decades of safety regulations and public awareness campaigns. The problem is not a lack of data. The federal government has been collecting detailed crash records for over 50 years through NHTSA FARS. What has been missing is a way to use that data to answer a more practical question: what combination of factors makes a crash more likely to kill multiple people rather than just one?

Our project looks at that gap. Instead of just describing where crashes happen or counting fatalities by state, we built predictive models to figure out which conditions at the time of a crash are linked to higher fatality counts. This is a regression task. We predicted FATALS, the number of deaths per crash, using predictors from crash timing, road environment, lighting, weather, crash type, and driver impairment.

The motivation is simple. If a city transportation planner knows that rural arterial roads at night with an impaired driver tend to have higher expected fatality counts, they can put enforcement resources and emergency response there. If a rideshare company knows that late-night weekend hours in certain road categories are linked to more severe crashes, they can think about how to get more drivers on the road at those times.

Our question is: given what we know about a crash as it is happening, what does the data say about how deadly it is likely to be?

3. Data Source

We used two files from the 2023 NHTSA Fatality Analysis Reporting System (FARS), which tracks every fatal car crash on U.S. public roads. The data comes from police reports, death certificates, and medical records — so it's quite reliable compared to survey.

The main file, `accident.csv`, has one row per crash and covers 37,654 fatal crashes across 80 variables — things like time of day, location, road type, weather, and crash type. The second file, `drimpair.csv`, tracks the impairment status of each driver involved. We combined the two files using a shared crash ID so we could add an impaired driver flag to the main dataset.

Detail	Value
Primary File	<code>accident.csv</code>
Secondary File	<code>drimpair.csv</code>
Total Crash Records	37,654
Original Features	80 (accident) + 6 (drimpair)
Working Features After Selection	14 predictors + 1 target
Unique Crash Identifier	ST_CASE
Data Source	NHTSA FARS 2023
Geographic Coverage	All 50 U.S. States

Table 1. Dataset Overview

After merging, our working dataset had 37,654 rows and 15 columns: 14 predictors and the target variable FATALS. No records were lost in the merge. Each row represents one unique fatal crash.

4. Data Investigation

4.1 Target Variable

Our target variable, FATALS, is the number of people killed in each crash. Since every crash in FARS resulted in at least one death, the minimum value is 1. The distribution is heavily right-skewed: about 93% of crashes resulted in exactly one fatality. That skew is actually part of what makes this problem challenging. Multi-fatality crashes are rare, but they are exactly the ones that matter most for prevention.

Statistic	Value
Minimum	1
25th Percentile (Q1)	1
Median	1
Mean	1.09
75th Percentile (Q3)	1
Maximum	8
Standard Deviation	0.35

Table 2. FATALS Summary Statistics

4.2 Predictor Variables

We selected 14 predictors based on relevance to crash severity and data coverage. They fall into four groups: time variables (HOUR, DAY_WEEK, MONTH), road and location variables (RUR_URB, FUNC_SYS, REL_ROAD, TYP_INT), environmental variables (LGT_COND, WEATHER), crash detail variables (MAN_COLL, HARM_EV, PERSONS, VE_TOTAL), and the merged impairment flag (impaired_driver).

Variable	Type	Description
FATALS	Numeric (Target)	Number of fatalities per crash (1 to 8)
HOUR	Numeric	Hour of day the crash occurred (0 to 23)
DAY_WEEK	Categorical	Day of the week (1 = Sunday, 7 = Saturday)
MONTH	Categorical	Month the crash occurred (1 = Jan, 12 = Dec)
RUR_URB	Categorical	Rural vs. urban classification of crash location
LGT_COND	Categorical	Lighting condition at time of crash
WEATHER	Categorical	Weather condition at time of crash
FUNC_SYS	Categorical	Functional road classification
REL_ROAD	Categorical	Relation to roadway (on road, shoulder, median)
TYP_INT	Categorical	Type of intersection
MAN_COLL	Categorical	Manner of collision (rear-end, head-on, etc.)
HARM_EV	Categorical	First harmful event in the crash sequence
PERSONS	Numeric	Total persons involved in the crash

Variable	Type	Description
VE_TOTAL	Numeric	Total vehicles involved in the crash
impaired_driver	Binary (0/1)	1 if at least one driver was impaired, 0 otherwise

Table 3. Variable Descriptions

4.3 Patterns We Noticed During Exploration

A few things stood out when we first looked at the data. Crashes with more vehicles and more people tended to have higher fatality counts. Crashes involving an impaired driver had an average fatality count of 1.132, compared to 1.078 for crashes without one. Rural crashes and crashes in dark conditions also tended to be more severe, which makes sense given higher speeds and slower emergency response times. Late-night hours, particularly between 10pm and 3am, showed noticeably higher average fatalities per crash.

5. Data Cleansing

5.1 Handling NHTSA Unknown Codes

NHTSA FARS uses numeric codes to represent unknown or unreported values instead of leaving fields blank. For example, HOUR = 99 means the hour was not reported, and WEATHER = 98 or 99 means weather conditions were unknown. If we left those in, the model would treat 99 as a real hour of the day, which would completely distort the results. We replaced all known unknown codes with NA before doing anything else.

Variable	Unknown Codes Replaced	Rows Affected
HOUR	99	282 (0.7%)
LGT_COND	8, 9	264 (0.7%)
WEATHER	98, 99	1,457 (3.9%)
RUR_URB	6, 8, 9	316 (0.8%)
FUNC_SYS	96, 98, 99	182 (0.5%)
REL_ROAD	98, 99	106 (0.3%)
TYP_INT	98, 99	195 (0.5%)
MAN_COLL	98, 99	123 (0.3%)
HARM_EV	99	37 (0.1%)

Table 4. Unknown Code Treatment by Variable

5.2 Imputation

Since the missing rates were all under 4% per column, we chose to impute rather than drop rows. For continuous predictors (HOUR, PERSONS, VE_TOTAL) we used median imputation because those distributions are right-skewed and the median is less sensitive to extreme values. For categorical columns coded as integers we used mode imputation, filling missing values with the most common category. After imputation, there were zero missing values across all 15 columns.

5.3 Outlier Detection and Treatment

We applied the IQR method to the three continuous predictors (HOUR, PERSONS, VE_TOTAL) to identify and treat outliers. Values outside Q1 minus 1.5 times the IQR and Q3 plus 1.5 times the IQR were capped at those boundaries rather than removed. Capping keeps the full row count while preventing extreme values from pulling the regression coefficients in one direction. We did not apply outlier treatment to FATALS itself, because the high values in the target represent real crashes with multiple deaths that the model needs to learn from.

5.4 Factor Conversion

Variables like DAY_WEEK, MONTH, RUR_URB, LGT_COND, WEATHER, FUNC_SYS, REL_ROAD, TYP_INT, MAN_COLL, HARM_EV, and impaired_driver are stored as integers in the raw data but are fundamentally categorical. A difference of 1 between two category codes does not mean anything numerically. We converted all of these to factors in R so the regression and decision tree models handle them as categories and create dummy variables automatically.

6. Feature Selection

To figure out which variables to include in the linear regression model, we used a method called stepwise selection. It tests adding or removing variables one at a time and keeps the ones that actually improve the model. It stops when nothing left to add or remove makes things better.

We ran it three ways — starting with all variables and cutting down, starting with none and building up, and a mix of both. All three approaches ended up picking the same variables, which is a good sign that those variables genuinely belong in the model. We also ran a quick check to make sure none of the variables were too closely related to each other, since that can cause problems and no issues came up.

The variables that made the final cut were: PERSONS, VE_TOTAL, impaired_driver, RUR_URB, LGT_COND, FUNC_SYS, MAN_COLL, HARM_EV, DAY_WEEK, and MONTH. The ones that got dropped — WEATHER, REL_ROAD, TYP_INT, and HOUR — just weren't adding enough to be worth keeping once the other variables were already in the model.

7. Model Development and Evaluation

7.1 Train / Validation Split

We split the dataset into a training set (70%, around 26,358 rows) and a validation set (30%, around 11,296 rows) using a random sample with `set.seed(123)` so the results are reproducible. The training set was used only for fitting the models and running stepwise selection. The validation set was held out completely and used only at the end to check how well each model generalizes to new data.

7.2 Model 1: Multiple Linear Regression

Our first model is a multiple linear regression predicting FATALS from the predictors selected through stepwise AIC. Linear regression estimates a coefficient for each predictor that represents its average contribution to expected fatalities when all other predictors are held constant. We used the `lm()` function in R, which we worked with throughout the course.

The model was fit on the training data and predictions were made for both sets. We evaluated performance using four metrics: Mean Error (ME), Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). We focused mainly on RMSE because it is in the same units as FATALS and penalizes larger errors more than MAE does. We also ran 5-fold cross-validation on the full dataset to get a more stable performance estimate.

Metric	Training Set	Validation Set	5-Fold CV
ME (Mean Error)	0.00	0.00	—
MAE (Mean Absolute Error)	0.1499	0.1505	0.1507
MSE (Mean Squared Error)	0.1092	0.1202	—
RMSE (Root Mean Squared Error)	0.3305	0.3467	0.3356

Table 5. Linear Regression Performance Metrics

The training and validation errors are very close to each other, so the model is not overfitting. The 5-fold cross-validation RMSE of 0.3356 lines up well with the holdout RMSE of 0.3467, which tells us the model is stable. An RMSE of around 0.35 means the model is off by about one-third of a fatality on average. Given that 93% of crashes have exactly one fatality and there is not much variation in the target, that is reasonable.

7.3 Model 2: Decision Tree (Regression)

Our second model is a CART regression decision tree using the `rpart` package with `method = anova`. Decision trees split the data recursively on the predictor that best reduces prediction error at each step. They do not assume a linear relationship between predictors and the target, which makes them a good comparison point for linear regression.

We grew an initial deep tree using a small complexity parameter (`cp = 0.001`) to explore many possible splits, then pruned it back to the optimal complexity by choosing the `cp` that minimized cross-validated error in the `cptable`. The pruned tree was plotted using `rpart.plot()` to check the splitting rules visually.

Metric	Training Set	Validation Set
ME (Mean Error)	0.00	0.00
MAE (Mean Absolute Error)	0.1403	0.1416

Metric	Training Set	Validation Set
MSE (Mean Squared Error)	0.1066	0.1195
RMSE (Root Mean Squared Error)	0.3265	0.3457

Table 6. Decision Tree Performance Metrics

The Decision Tree's validation RMSE of 0.3457 is very close to the linear regression result, and the gap between training and validation errors is small, which means pruning worked well. The top splits in the pruned tree were driven by PERSONS, VE_TOTAL, and impaired_driver, which is the same pattern stepwise selection identified in the linear model.

7.4 Model Comparison

Model	Val. MAE	Val. RMSE	Overfitting?
Linear Regression (Stepwise)	0.1505	0.3467	No — train and validation nearly identical
Decision Tree (Pruned)	0.1416	0.3457	Minimal — well controlled by pruning

Table 7. Model Comparison Summary

Both models perform very similarly. The Decision Tree is slightly better on RMSE (0.3457 vs. 0.3467) and MAE (0.1416 vs. 0.1505), but the gap is small enough that either could work in practice. We consider the Linear Regression our primary model because the coefficients give a clear, interpretable picture of how each predictor contributes to expected fatalities. The Decision Tree works as a strong secondary model that confirms the same variables matter.

8. Practical Implications

The model results are not just numbers on a page. They point to specific situations where safety resources can make a real difference.

8.1 Impaired Drivers Make Crashes More Deadly

The `impaired_driver` flag was selected as a significant predictor in both models. Crashes with at least one impaired driver had higher predicted fatality counts than similar crashes without one. This supports placing breathalyzer kiosks at bars and working with rideshare companies on late-night coverage. It is a practical intervention with a measurable connection to crash severity in the data.

8.2 Number of Persons and Vehicles Are the Strongest Predictors

`PERSONS` and `VE_TOTAL` were the two strongest predictors across both models. More people and more vehicles in a crash means more potential victims. This points to multi-vehicle scenarios like highway pileups and crashes involving large trucks as a distinct risk category. Emergency management agencies could use crash type and road classification data to pre-position resources near segments with historically high multi-vehicle counts.

8.3 Rural Roads at Night Are the Highest-Risk Environment

Both `RUR_URB` and `LGT_COND` were retained in the final model, and their coefficients show that rural crashes in dark conditions are linked to higher expected fatality counts. Rural road crashes tend to be more severe because of higher speeds, fewer barriers, and slower emergency response. Investing in rural road lighting and better nighttime dispatch coverage on arterial roads makes sense given what the data shows.

8.4 Seasonal and Day-of-Week Patterns

`DAY_WEEK` and `MONTH` both stayed in the stepwise-selected model, which means timing matters beyond just the hour of day. Weekend crashes and summer month crashes tended to have higher predicted fatality counts, which lines up with more recreational driving and more alcohol involvement during those periods. Concentrating sobriety checkpoints and safety campaigns on Friday and Saturday nights and during summer holiday weekends makes more sense than spreading them evenly across the year.

9. Conclusion and Recommendations

This project set out to answer a practical question: can we use crash-level data to predict how many people will die in a fatal crash? Using 37,654 records from the 2023 NHTSA FARS dataset and a merged driver impairment file, we built two regression models that both achieved validation RMSEs around 0.35 with no meaningful overfitting.

The most consistent finding across both models is that crash severity is driven by a mix of factors working together: how many people and vehicles are involved, whether a driver was impaired, what kind of road it happened on, and what the lighting conditions were. The models let us quantify how those things combine, and both approaches agreed on the same predictors, which gives us confidence that the results are real.

Based on the analysis, here are our recommendations:

1. State transportation agencies should prioritize lighting improvements and median barriers on rural arterial roads, which the models flagged as the highest-risk environment for multi-fatality crashes.
2. Public health departments and law enforcement should increase sobriety checkpoint operations on Friday and Saturday nights during summer months, when the models predict the highest expected fatality counts.

3. Rideshare companies should increase driver availability between 10pm and 3am on weekend nights to reduce the number of people who choose to drive while impaired.
4. Bar and restaurant associations can work with local governments to install breathalyzer kiosks at high-volume venues, with priority given to rural counties where transportation alternatives are limited.
5. Emergency management agencies should build response plans for road segments that combine rural classification, arterial road type, and high historical PERSONS counts, since that combination came up consistently in both models.

For future work, adding real-time traffic volume data or building a classification model that flags crashes likely to exceed a fatality threshold could both add value. Reframing the problem as binary classification, single fatality versus multiple fatalities, is also worth exploring since the skewed distribution of FATALS makes regression more challenging. The main takeaway from this analysis is that the deadliest crashes are not random. They cluster around predictable conditions, and that means there is a real opportunity to act on them.

References

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Appendix

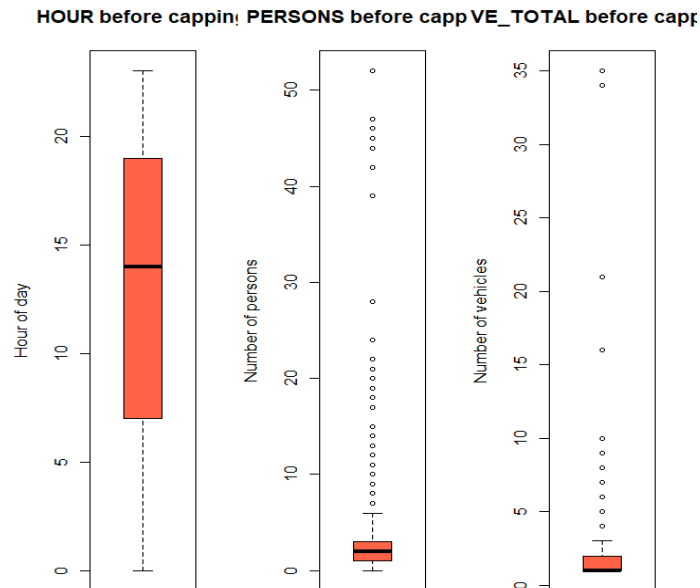


Image 1.1 – Boxplots before capping

Same three variables, but before any cleaning. You can see some crashes had wildly high values (like 50+ persons in one crash), which are not realistic and would've messed up the models.

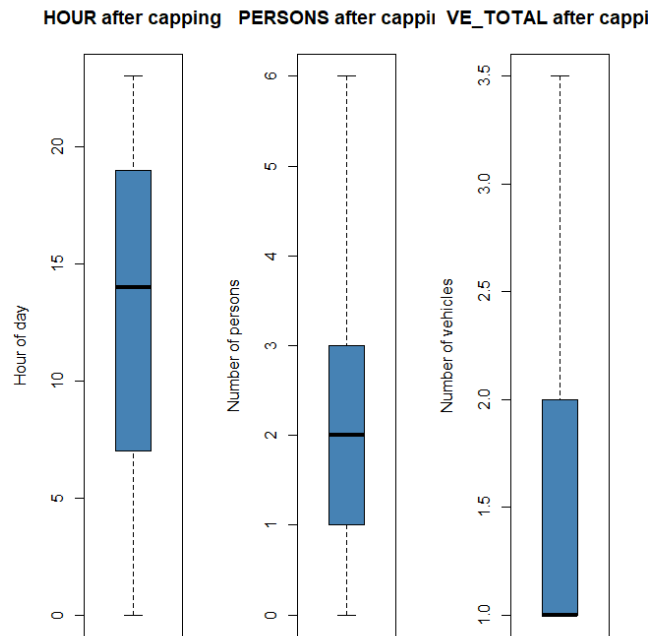


Image 1.2 – Boxplots after capping

These show HOUR, PERSONS, and VE_TOTAL after extreme values were cleaned up. Everything looks much more contained compared to before — no crazy outliers pulling the data out of shape.

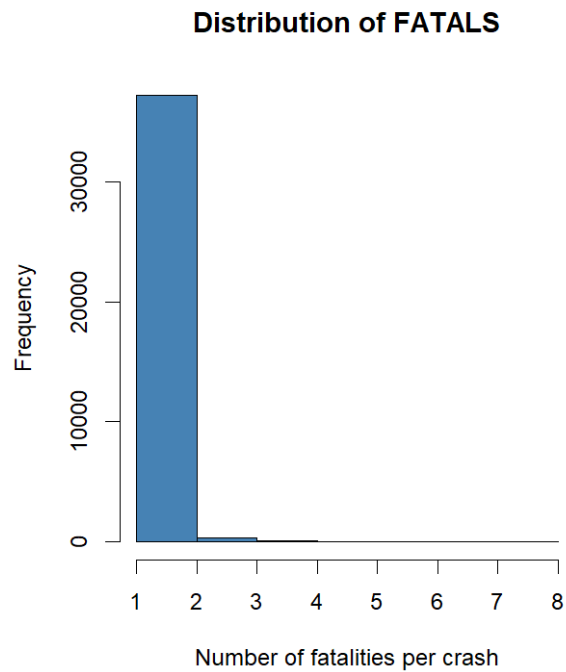


Image 1.3 – Distribution of FATALS

Shows how many fatalities occurred per crash. Almost every crash had just 1 fatality — crashes with 2 or more are pretty rare. This is the variable the models are trying to predict.

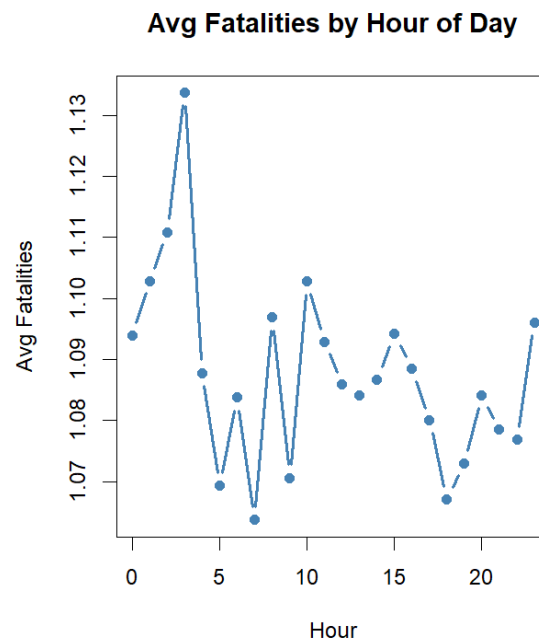


Image 1.4 – Avg fatalities by hour of day

Looks at whether the time of day affects fatality count. There's a small bump around 2–3 AM, but honestly the differences are tiny — we're talking about a range of roughly 1.07 to 1.13 on average.

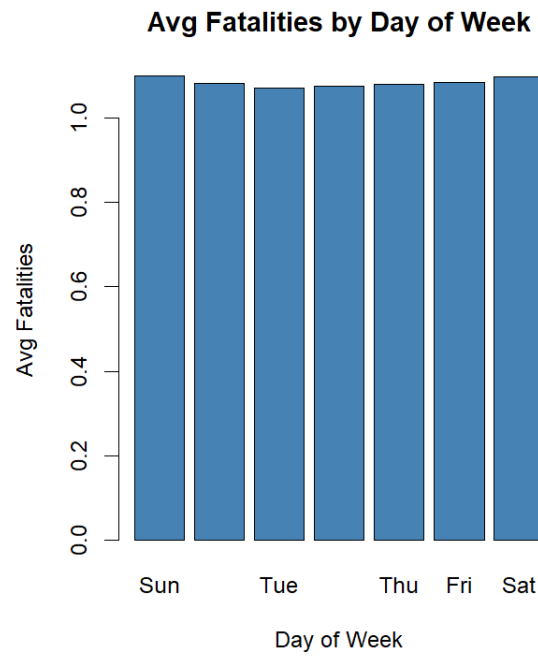


Image 1.5 – Average fatalities by day of week

Same idea but by day of the week. All the bars are basically the same height, so day of week doesn't seem to matter much for predicting fatalities.

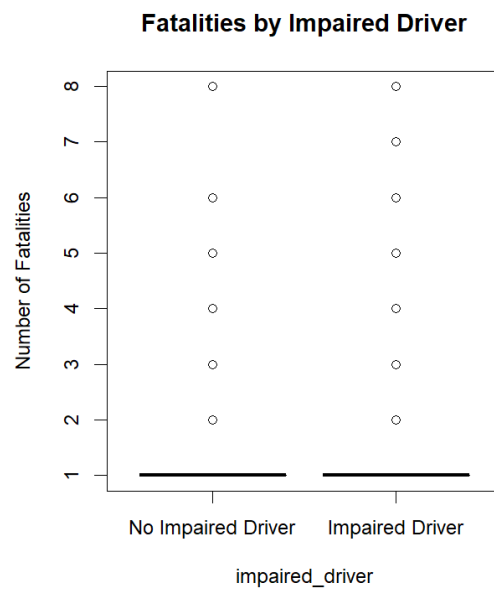


Image 1.6 – Fatalities by impaired driver

Compares crashes where a driver was impaired vs. not. Both groups look pretty similar, with most crashes having just 1 fatality either way.

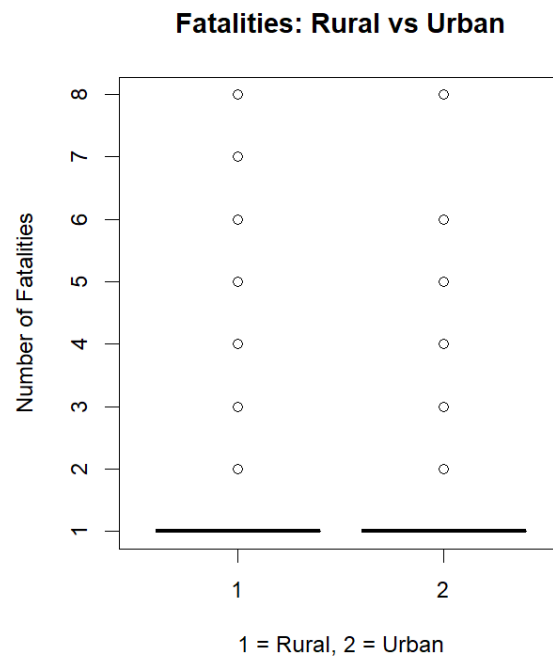


Image 1.7 – Fatalities: Rural vs Urban

Compares rural and urban crashes. Again, both groups look nearly identical in terms of fatality counts — not a huge difference between the two settings.

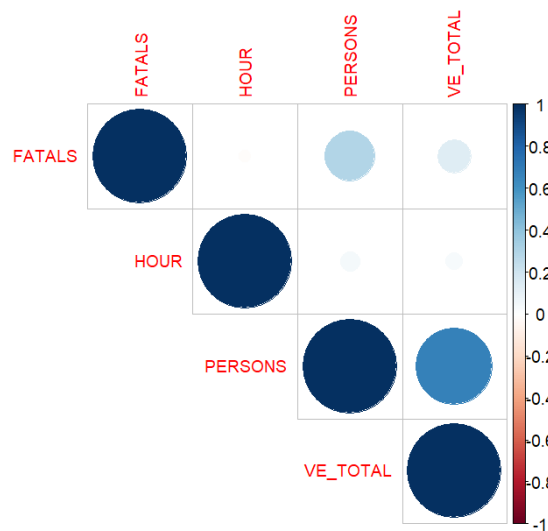
Correlation - Numeric Predictors vs FATALS

Image 1.8 – Correlation: Numeric Predictors vs FATALS

Shows how related the numeric variables are to each other and to fatalities. PERSONS and VE_TOTAL are somewhat related to each other, but none of the predictors are strongly linked to fatality count, which means predicting it accurately is a bit of a challenge.